

Mini ERP

Version 2.0 · April 2026

Workforce & HR

Payroll

Accounts Payable

Expenses

Procurement

General Ledger

AI / MLOps

Finance Chat ★ New

Administration

End-User Guide

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Application Overview

Mini ERP is a unified enterprise resource planning platform covering HR, payroll, finance, procurement, and AI-driven insights in a single web application. Phase 3 adds **Finance Chat** -- natural-language queries over all ERP data.

Workforce Management	Manage employees, departments, job titles, and employment lifecycle from hiring to termination.
Payroll	Process pay periods, generate payslips, manage pay groups, and track gross/net compensation with deductions.
Accounts Payable	Track vendors, manage invoices through full approval lifecycle (draft → approved → paid), and reconcile payments.
Expenses	Submit and approve employee expense reports with multi-status workflow -- draft, submitted, approved, rejected, paid.
Procurement	Raise purchase requisitions, convert them to purchase orders, and track delivery status through goods receipt.
General Ledger	Maintain the chart of accounts, post journal entries, manage fiscal periods, and view the trial balance.
AI / MLOps	Machine-learning models surface attrition risk, expense violations, payroll anomalies, and invoice mis-classifications in real time.
Finance Chat ★	Ask questions about your ERP data in plain English -- the AI translates them to SQL, queries live data, and returns a formatted answer. New in Phase 3.
Administration	Manage platform users, assign roles, and control access permissions across every module.

Technology Stack

Layer	Technology
Backend	FastAPI · Python 3.11 · SQLAlchemy 2.0 (async) · Alembic
Database	PostgreSQL 15 · pgvector (Phase 3 embeddings)
Cache / Queue	Redis 7 · Celery
ML / LLM	scikit-learn · joblib · Anthropic Claude · Groq · Ollama
Frontend	React 18 · TypeScript · Vite · Tailwind CSS
Infrastructure	Docker · Docker Compose · GCP free tier · Nginx

User Roles & Access

Role	Username	Access Level
Platform Admin	platform_admin	Full access to all modules + administration
HR Admin	hr_admin	Full HR, workforce, and payroll access
Finance Admin	finance_admin	Full AP, expenses, GL, and procurement access
Workforce User	workforce_user1	Read-only workforce module
Payroll User	payroll_user1	Read-only payroll module
AP User	ap_user1	Read-only accounts payable
GL User	gl_user1	Read-only general ledger

Getting Started

Follow these steps to begin using Mini ERP for the first time.

1 Open the application	Navigate to the Mini ERP URL in your web browser (Chrome or Edge recommended). The application is hosted at http://34.13.57.203:3000 .
2 Sign in with your credentials	Enter your username and password on the Login page. Your account will have been set up by the Platform Administrator.
3 Explore the Dashboard	After login you land on the Dashboard, which shows high-level counts across employees, invoices, expenses, and purchase orders.
4 Navigate using the sidebar	The collapsible left sidebar groups all modules. Click any menu item to jump to that section. Your role determines which modules are accessible.
5 Use search and filters	Most list pages include a search bar and status filter dropdown. Type to search by name, number, or email.
6 Create & edit records (Admin only)	Users with Admin roles see action buttons and pencil edit icons. Standard users have read-only access.
7 Try Finance Chat ★	Click AI Chat in the sidebar to open Finance Chat. Type any question about your financial data in plain English and press Enter. No SQL knowledge required.

Default Login Credentials (Demo)

Role	Credentials
Platform Administrator	platform_admin / Admin@123!
HR Administrator	hr_admin / Admin@123!
Finance Administrator	finance_admin / Admin@123!
Module-specific users	workforce_user1, payroll_user1, etc. / User@123!

SECTION 1

Login Page

/login

Sign in to Mini ERP and securely access your organisation's data

- Enter your username and password then click **Sign in**
- Invalid credentials show an inline error message
- After successful login you are redirected to the Dashboard
- Sessions are maintained via secure JWT tokens stored in the browser
- Each user's access is governed by their assigned role
- Contact your Platform Admin if you are locked out

Tip: Use platform_admin / Admin@123! for full admin access in the demo environment.

SECTION 2

Dashboard

/dashboard

Your at-a-glance view of key ERP metrics

- **Total Employees** -- active headcount from the workforce module
- **AP Invoices** -- total invoices across all statuses
- **Expense Reports** -- all expense reports in the system
- **Purchase Orders** -- total POs raised with vendors
- Getting Started panel with quick-links to commonly used pages
- Data refreshes automatically on each visit

SECTION 3

Employees

/workforce/employees

Manage employees, departments, and organisational structure

- Search employees by name or email in real-time
- Attrition risk badge -- green (<40%), amber (40–60%), red (>60%)
- Status badges: active, inactive, on leave, terminated
- Paginated table -- 20 records per page with navigation
- **Admin:** Add a new employee via the **Add Employee** button
- **Admin:** Edit department, job title, or status via the pencil icon

Note: Attrition Risk scores are generated by the **attrition_predictor** ML model. An employee showing >60% risk should be flagged for a retention conversation. If all scores show --, go to AI / MLOps → Model Registry and train the model first.

SECTION 4

Departments

/workforce/departments

Manage employees, departments, and organisational structure

- View all departments with headcount and manager details
- Department codes are used as reference identifiers in other modules
- Cost centre linking for GL expense allocation
- **Admin:** Create new departments and assign managers
- **Admin:** Edit department name, code, or manager
- Changes to departments are reflected immediately in the Employees page

SECTION 5

Pay Periods

/payroll/periods

Manage pay periods, process payroll runs, and view employee payslips

- View all pay periods with start date, end date, and status
- Statuses: open, closed, processed, approved
- Each period is linked to a pay group (e.g. Monthly Staff)
- **Admin:** Create new pay periods for a pay group
- **Admin:** Close or process a period to trigger payslip generation
- Closed periods are locked to prevent retroactive edits

SECTION 6

Payslips

/payroll/payslips

Manage pay periods, process payroll runs, and view employee payslips

- Employee name linked to their HR profile
- Pay period name for quick identification
- Gross Pay shown in standard currency format
- Deductions displayed in red for quick identification
- Net Pay highlighted in green -- the take-home amount
- Status badge: draft, processed, approved, paid

SECTION 7

Vendors

/ap/vendors

Manage vendors, invoices, and outgoing payments

- View vendor name, code, and contact information
- Payment terms stored per vendor (e.g. Net 30, Net 60)
- Active / inactive status for controlling whether new orders can be raised
- Vendor currency and bank details for payment processing
- **Admin:** Register new vendors with the **Add Vendor** button
- **Admin:** Update vendor details or mark a vendor inactive

SECTION 8

Invoices

/ap/invoices

Manage vendors, invoices, and outgoing payments

- Filter invoices by status: All, draft, submitted, approved, paid, overdue, cancelled
- Invoice number, vendor, issue date, and due date at a glance
- Total Amount vs Paid Amount columns to track outstanding balance
- Overdue invoices flagged automatically past due date
- **Admin:** Create new invoices -- select vendor, dates, amounts, and description
- **Admin:** Update invoice status to advance through the approval workflow

Invoice Status Reference

Status	Meaning	Outstanding?
DRAFT	Created but not yet submitted	No
SUBMITTED	Submitted for review	Yes
UNDER_REVIEW	Being reviewed by finance team	Yes
APPROVED	Approved, awaiting payment	Yes
MATCHED	Matched against a purchase order	Yes
POSTED	Posted to the general ledger	Yes
PAID	Payment has been made	No
REJECTED	Rejected -- will not be paid	No
CANCELLED	Cancelled by requester or admin	No

Outstanding balance = invoices with status SUBMITTED, UNDER_REVIEW, APPROVED, MATCHED, or POSTED. You can also ask Finance Chat: *What is the total AP outstanding balance?*

Once an invoice is marked PAID or CANCELLED it cannot be reversed through the UI. Contact your Finance Administrator for corrections.

SECTION 9

Expense Reports

/expenses/reports

Submit, review, and approve employee expense claims

- Filter by status: draft, submitted, approved, rejected, paid
- Report title, submitting employee, total amount, and submission date visible
- Status badge tracks position in the approval workflow
- **Admin:** Approve a submitted report with the green tick button
- **Admin:** Reject a report -- a reason must be provided
- **Admin:** Create new expense reports on behalf of employees

Note: The AI module can flag expense lines with a violation risk >60% based on policy rules and historical patterns. Check AI / MLOps → Insights → Expense Violation Risk before approving high-value reports.

SECTION 10

Purchase Requisitions

/procurement/requisitions

Raise purchase requisitions and manage purchase orders

- View all PRs with requester, department, and requested amount
- Statuses: draft, submitted, approved, rejected, converted to PO
- Justification and required-by date captured per requisition
- **Admin:** Approve or reject submitted requisitions
- **Admin:** Create new requisitions on behalf of a department
- Approved PRs can be converted directly into a Purchase Order

SECTION 11

Purchase Orders

/procurement/orders

Raise purchase requisitions and manage purchase orders

- Filter by status: draft, sent, acknowledged, partially received, received, cancelled
- PO number, vendor, order date, expected delivery, and total amount
- Track delivery progress from sent → acknowledged → received
- **Admin:** Create new POs linked to a vendor with order details
- **Admin:** Update PO status and adjust expected delivery date
- Cancelled POs are retained for audit trail purposes

SECTION 12

Chart of Accounts

/gl/accounts

Chart of accounts, journal entries, and financial reporting

- Account types: asset, liability, equity, revenue, expense -- colour-coded by badge
- Account codes follow standard accounting numbering conventions
- Normal balance indicated (debit/credit) per account type
- Active / inactive status controls whether the account accepts postings
- **Admin:** Add new accounts with code, name, type, and normal balance
- **Admin:** Edit or deactivate accounts (accounts with postings cannot be deleted)

SECTION 13

Journal Entries

/gl/journals

Chart of accounts, journal entries, and financial reporting

- Journal reference number, description, and posting date
- Source indicates whether the journal was auto-generated or manually created
- Total debit and credit amounts -- must balance for a valid journal
- Status: draft journals are editable; posted journals are locked
- **Admin:** Create manual journal entries with multiple debit/credit lines
- **Admin:** Post a draft journal to make it permanent in the ledger

Important: Posted journals cannot be edited or reversed through the UI. Any correction requires creating a reversing journal entry. Contact your Finance Administrator.

SECTION 14

Trial Balance

[/gl/trial-balance](#)

Chart of accounts, journal entries, and financial reporting

- All active GL accounts listed with total debits and credits
- Account type colour-coded: asset (blue), liability (red), equity (purple), revenue (green), expense (orange)
- Totals row at the bottom showing overall debit and credit sums
- **Balanced** (green) badge when debits = credits; **Out of Balance** (red) if discrepancy exists
- Read-only report -- no editing from this view
- Refreshed automatically from the latest posted journal entries

SECTION 9

AI Insights Tab

/ai → Insights

Machine-learning powered insights, model registry, and prediction audit log

- **Attrition Risk** -- top 10 employees most likely to leave
- **Expense Violation Risk** -- top 10 expense lines suspected of policy breach
- **Invoice Classifications** -- AI-predicted spend category vs actual
- **Payroll Anomalies** -- top 10 payslips with statistically unusual pay
- Risk bar shows colour-coded percentage (green <40%, amber 40–60%, red >60%)
- **View All >60%** button expands to a full modal of high-risk records

Note: If a section shows Model not trained, go to the Model Registry tab and train the relevant model first. Models must be trained before predictions are available.

SECTION 9

Model Registry Tab

[/ai](#) → *Model Registry*

Machine-learning powered insights, model registry, and prediction audit log

- **Attrition Predictor** -- predicts employee attrition risk using HR features
- **Expense Violation** -- detects expense lines that breach policy or norms
- **Payroll Anomaly** -- identifies statistically unusual payslips
- **Invoice Classifier** -- classifies invoices into spend categories via TF-IDF
- Metrics displayed: Accuracy, ROC-AUC, F1 Score, CV-AUC (cross-validation)
- Use the Refresh button to reload model status from the server

SECTION 9

Prediction Log Tab

[/ai](#) → *Prediction Log*

Machine-learning powered insights, model registry, and prediction audit log

- Model ID (shortened UUID) identifies which model made the prediction
- Entity type/ID shows what record was scored (employee, payslip, expense, invoice)
- Prediction label: high_risk, violation, anomalous, or low_risk / normal
- Probability percentage -- the model's confidence in the prediction
- Latency in milliseconds for performance monitoring
- Paginated -- 20 rows per page; newest predictions first

SECTION 9

Finance Chat (AI Chat) ★ New in v2.0

/chat

Ask your ERP data questions in plain English

Finance Chat is a natural-language interface to all ERP data. Type a question in plain English, and the AI automatically generates and executes the correct SQL query, then presents the result in a clear, formatted answer. No SQL knowledge required.

How it works -- end to end

Step	What happens
1 User types a question	Natural language question entered in the chat input and submitted with Enter
2 LLM receives schema context	The full ERP database schema + invoice status definitions are sent to the LLM as context
3 LLM generates SQL	The LLM calls the run_sql tool with a validated SELECT query
4 Backend executes query	Query is safety-checked, wrapped with a row limit, and run against PostgreSQL in a savepoint
5 Results formatted	Query results are converted to a markdown table and fed back to the LLM
6 LLM answers in plain English	The LLM produces a natural-language summary -- never raw JSON
7 Response displayed	Answer shown in the chat; a collapsible SQL panel shows the generated query + data

Example questions you can ask

Question	What it queries
What is the total AP outstanding balance?	SUM of invoice total_amount where status NOT IN (PAID, REJECTED, CANCELLED)
Show department-wise AP outstanding balance	Groups outstanding invoices by department via GL cost centre
Which employees have the highest attrition risk?	Orders hcm.employees by attrition_risk_score DESC
Top 5 vendors by invoice amount this year	Groups ap.invoices by vendor, filtered by invoice_date year
List all invoices flagged as duplicates	Filters ap.invoices WHERE is_duplicate = true
Which expense reports are pending approval?	Filters expenses.expense_reports WHERE status = SUBMITTED
Budget vs actual spend by department	Joins gl.budgets with hcm.departments
What is our payroll cost this month?	SUM payroll.payslips.gross_pay for current pay period

SECTION 9 (continued)

Finance Chat -- Using the Interface

Opening the chat

- Click **AI Chat** in the left sidebar to navigate to `/chat`
- The welcome screen displays 6 example prompts -- click any to start immediately
- The text input is always enabled -- just start typing and press **Enter**
- A new conversation is created automatically when you send your first message

During a conversation

- Each assistant reply shows the answer in plain English
- Click the **SQL Query** panel below any answer to expand the generated SQL and raw data table
- Latency (ms) and token counts are shown per message
- Press **Shift+Enter** to add a newline without sending

Managing conversations

- Past conversations are listed in the left panel, ordered by most-recent message
- Conversation titles are auto-generated from your first question
- Click the trash icon on a conversation to archive it
- Click **New conversation** to start a fresh chat session

LLM Provider configuration (Admin)

Provider	Config key	Notes
Anthropic Claude	LLM_PROVIDER=anthropic	Default. Best quality. Requires ANTHROPIC_API_KEY.
Groq	LLM_PROVIDER=groq	Free tier. Fast. Requires GROQ_API_KEY.
Ollama (local)	LLM_PROVIDER=ollama	No API key. Requires local Ollama instance.

Security & data access

- Finance Chat is **read-only** -- it can only generate SELECT queries
- Forbidden keywords (INSERT, UPDATE, DELETE, DROP, etc.) are blocked server-side
- All queries are wrapped in a row limit (default 500 rows) to prevent data dumps
- Full audit trail: every conversation and message is persisted to the database
- Access is controlled by the AI module permission in RBAC -- Finance Admin and above

Requires ANTHROPIC_API_KEY (or GROQ_API_KEY / local Ollama) to be configured in the backend environment. Contact your Platform Administrator if Finance Chat shows a 503 error.

Results are generated by an AI model and may occasionally be incorrect for very complex queries. Always verify critical financial figures against the source module (e.g. AP → Invoices) before using them in reports.

SECTION 10

User Management

/admin/users

Manage platform users, roles, and access permissions

- View all platform users with username, email, and assigned role
- Account status: active or inactive shown per user
- Role badge indicates the user's permission level
- **Admin:** Create new user accounts and assign them a role
- **Admin:** Deactivate user accounts to revoke access without deleting data
- **Admin:** Reassign roles to adjust a user's module permissions

Note: Only the Platform Administrator (platform_admin) can access this page. If you need a new user account or a role change, contact your Platform Administrator.

Appendix

A Phase Roadmap

Phase	Description	Status
Week 0	GCP setup, CI/CD, DB schema, RBAC	■ Done
Phase 1	MVP ERP -- Workforce, Payroll, AP, Expenses, Procurement, GL	■ Done
Phase 2	Machine Learning -- Attrition, Expense Violations, Payroll Anomaly, Invoice Classifier	■ Done
Phase 3	RAG + LLM -- Finance Chat, Duplicate Invoice Detection, OCR Pipeline	■ In Progress
Phase 4	Deep Learning -- LSTM Budget Forecaster, CNN Invoice Classifier	■ Pending
Phase 5	AI Agents -- Invoice Agent, Expense Audit Agent, Onboarding Agent	■ Pending
Phase 6	Agentic Networks -- Financial Close Network, Workforce Planning Network	■ Pending
Week 11	Polish, Docs, Portfolio Deploy	■ Pending

B Version Changelog

Version	Date	Summary
v2.0.0	Apr 2026	Phase 3: Finance Chat (Text-to-SQL), Duplicate Invoice Detection, OCR Pipeline, pgvector
v2.0.1	Apr 2026	Bug fixes: savepoint isolation, inline tool-call detection, AP status enum context, frozen chat input
v1.0.0	Feb 2026	Phase 1 MVP: full ERP CRUD across 7 modules + RBAC
v1.1.0	Mar 2026	Phase 2 ML: 4 ML models, AI insights dashboard, model registry, prediction log

C Keyboard Shortcuts

Key	Action
Enter	Send message (Finance Chat)
Shift+Enter	New line without sending (Finance Chat)
Ctrl+/	Toggle sidebar collapse
Esc	Close modal / cancel action